Wahid Al Azad Navid

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Research Interests

Computational genomics: algorithms and probabilistic models for sequence analysis; machine learning for biological sequence data; multimodal learning and its interpretability.

Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering Jan 2022 – June 2026

- **CGPA: 3.96 / 4.00, ranked 11 of 128.** Dean's List in all years.
- Thesis: *kRISP-meR: A Reference-free Guide-RNA Design Tool for CRISPR/Cas9*. Supervisor: Dr. Atif Hasan Rahman.

Notre Dame College Dhaka, Bangladesh
HSC, Science 2018 – 2020

- GPA: 5.00 / 5.00

Research Experience

Undergraduate Thesis, BUET CSE, kRISP-meR: reference-free CRISPR/Cas9 guide-RNA design Dhaka, Bangladesh
May 2025 – June 2026

Supervised by Dr. Atif Hasan Rahman, in collaboration with Dr. Mahmudur Rahman Hera.

- Designs CRISPR/Cas9 guide RNAs directly from the raw sequencing reads of the individual being edited, without a reference genome or a genome assembly.
- Models the k-mer frequency spectrum of the reads as a Poisson mixture fitted with expectation-maximization to infer genomic copy number, and from it the expected number of cuts of every candidate guide.
- Benchmarked against CRISPOR and GuideScan2 on human, mouse, and yeast data; rejects guides that appear unique in the reference genome but are duplicated in the individual (CHM13 vs. GRCh38, N22 rice vs. Nipponbare).
- Compared with CIRCLE-seq cleavage data in K562 cells, using copy number estimated from ENCODE whole-genome sequencing reads; identified experimentally confirmed off-target sites present at more copies in the cell line than in the reference.

LimbicsAI, Software and ML Engineer (part-time) Remote
May 2025 – present

- Co-developed a multimodal pavement assessment model fusing roadway video, vehicle-mounted IMUs, and GPS (paper accepted for presentation at TRB 2027), and built the automated AWS pipeline that processes the field data end to end.
- Modelled ground-deformation displacement time series from Sentinel-1 InSAR data; implemented PSI and PCI (ASTM D6433) pavement condition scoring and integrated it into the product.
- Set up the LoRaWAN sensor data pipeline (ChirpStack, EMQX, InfluxDB) and a bridge scour monitoring test system, including the physical sensor setup.
- Wrote crawlers for roadway data collection; rebuilt the company web front end, integrated an AI co-pilot, and deployed the internal documentation server.

Conference Presentations and Preprints

A Multimodal Framework for Pavement Condition Assessment Using Roadway Imagery and Vehicle Response January 2027
Shams E Shifat, ***Wahid Al Azad Navid***, Ashrafur Rahman, AQM Zohuruzzaman, Zhao Gao, Nasif Shafi, Mosharaf Chowdhury, Sadik Khan
Accepted for presentation, Transportation Research Board (TRB) Annual Meeting

Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization 2026
H. M. Shadman Tabib, Istiak Ahmmed Rifti, …, ***Wahid Al Azad Navid***, et al.
arXiv:2602.14291

Manuscripts in Preparation

kRISP-meR: A Reference-free Guide-RNA Design Tool for CRISPR/Cas9 2026
Wahid Al Azad Navid, Fuad Ahmed Udoy, Mahmudur Rahman Hera, Atif Hasan Rahman
Benchmarking complete; manuscript being finalised

Disentangling Cross-Modal Information Pathways via Mechanistic Interpretability of Multimodal Fusion Models 2026
Abrar Rahman Abir, ***Wahid Al Azad Navid***, Sheikh Azizul Hakim

DocTalkBN: A Novel Dataset of Expert Telemedicine Conversations in Bengali 2026
Anik Saha, Mst. Fahmida Sultana Naznin, Sadatul Islam Sadi, Ananya Shahrin Promi, ***Wahid Al Azad Navid***, Rifat Shahriyar

Teaching

Presidency University, Lecturer, Department of Computer Science and Engineering — Dhaka, Bangladesh
June 2026 – present

- **Fall 2026:** Object Oriented Programming (theory and lab); Microprocessors and Embedded Systems.
- **Summer 2026:** Structured Programming Language (theory and lab); Digital Logic Design Lab.

Honors and Service

- **Dean's List**, BUET (2022 – 2026)
- **University Merit Scholarship**, BUET (2022 – 2026)
- **Best UI/UX Award**, Hackathon, BUET CSE Fest (2024)
- **Runners-up**, Project Showcasing, IUT Skill Innovation Fest (2024)
- Government Scholarships: HSC (2020), SSC (2018), JSC (2016)
- Regional Winner, Bangladesh Junior Science Olympiad (2018)
- **Lead Organiser**, BUET CSE Fest 2026: 7 events, 100+ universities; secured title and supporting sponsors
- Logistics Director, BUET Cyber Security Club (May 2025 – May 2026); regular capture-the-flag participant

Selected Projects

GradPilot July 2025
- ML-assisted platform for graduate applications (university matching, SOP review) built as microservices with CI/CD. Spring Boot, Next.js, MySQL, Docker, GitHub Actions.

DHCP Starvation and Spoofing Attack June 2025
- Simulated both attacks in an isolated lab, measured network disruption, and evaluated DHCP snooping and port security. Python, Scapy, ns-3.

4-bit MIPS Processor Feb 2024
- Designed the datapath and control unit for a custom instruction set and built it on physical ICs with an ATmega32A.

C Compiler Dec 2023
- Compiler for a subset of C with scoped symbol tables, type checking, and code generation. Flex, Bison.

Technical Skills

Programming: Python, C/C++, Java
ML and Bioinformatics: PyTorch, NumPy, Pandas, Matplotlib; Jellyfish, Bowtie2, Pilon
Systems and Cloud: AWS, Docker, Git, xv6, ns-3, LoRaWAN (ChirpStack, EMQX, InfluxDB)
Databases and Web: MySQL, PostgreSQL, MongoDB; Node.js, React, Next.js, Spring Boot